

Macroeconomics III  
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Chapter 11

EFFICIENCY  
WAGES AND  
UNEMPLOYMENT

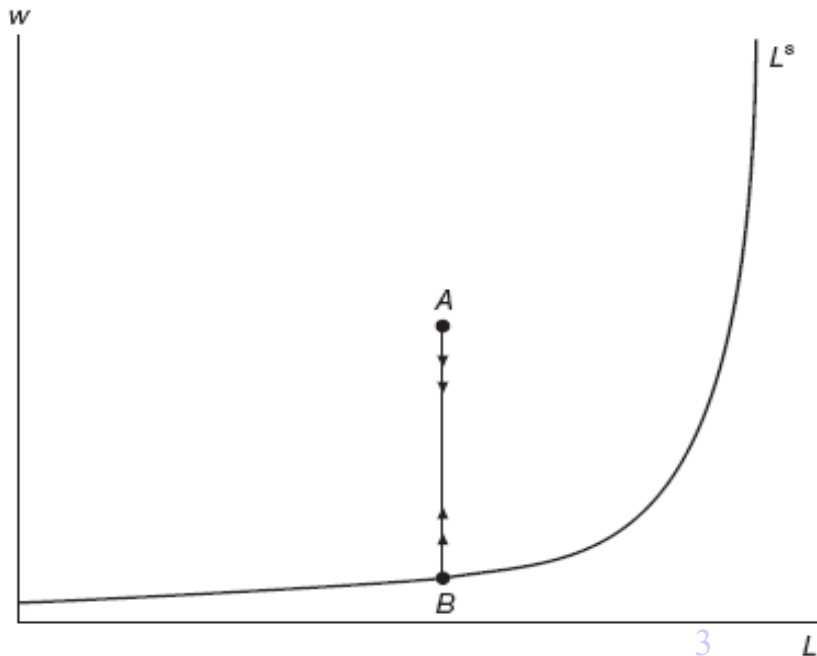
# EFFICIENCY WAGES

A long-run, real wage rigidity arising from a minimal departure from a traditional competitive labour market:

- Many buyers (firms) and sellers (workers), labour is homogenous, that is, the conditions normally assumed to imply perfectly competitive wage adjustment.
- But also, labour productivity is assumed to depend positively on (real) wages: **efficiency wages**. An example of quality depending on price.
- There are good reasons for this assumption and it has important macroeconomic implications, as we shall see.
- First we give an illustration of the implications of efficiency wages in...

# A SIMPLE PARTIAL EQUILIBRIUM MODEL

- Consider a firm with revenue function:  
 $zR(L)$ ,  $R'(L) > 0$ ,  $R''(L) < 0$ , where  $z$  is supply/demand shock
- For a clean argument we first assume that the firm is a monopsonist on the labour market and decides the real wage,  $w$ , and employment,  $L$ .
- Labour supply curve:  $L^s(w)$ . Restriction:  $L \leq L^s(w)$ .  
Profits:  $zR(L) - wL$ .



- The optimal  $(w, L)$  is **on** the labour supply curve: there is no unemployment!

- This argument assumes that labour productivity does not depend on pay.
- Assuming productivity/efficiency =  $a(w)$ ,  $a' > 0$ , implies a new revenue function:

$$zR(a(w)L)$$

and:

$$\text{profits} = zR(a(w)L) - wL$$

- The argument for the firm *necessarily* choosing a  $(w, L)$  on the labour supply curve breaks down.
- Is a full optimum of the firm compatible with a  $(w, L)$  above the labour supply curve?

- The firm chooses  $w$  and  $L$  to

$$\max zR(a(w)L) - wL \quad \text{st. } L \leq L^s(w).$$

- The first-order conditions, disregarding the constraint,  $L \leq L^s(w)$ , are:

$$zR'(a(w)L)a'(w)L = L,$$

$$zR'(a(w)L)a(w) = w.$$

- From the second equation we get  $zR'(a(w)L) = w/a(w)$ . Inserting this into the first gives:

$$\frac{a'(w)w}{a(w)} = 1.$$

- This is the famous "Solow condition"! Intuition:

- **Without** efficiency wages: profits =  $zR(L) - wL$ . No matter what the optimal level of  $L$  is, set  $w$  as small as possible.
- **With** efficiency wages:

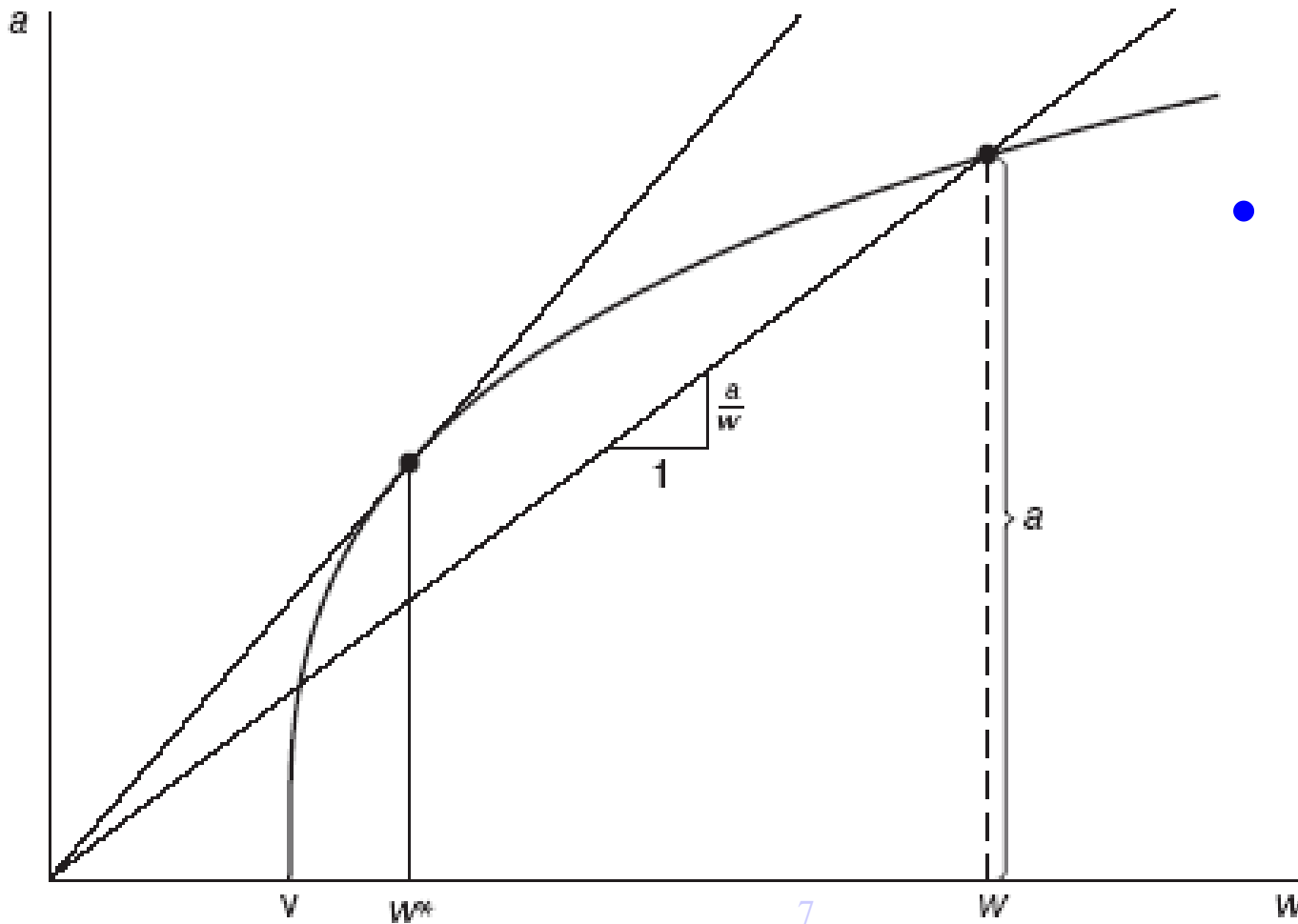
$$\text{profits} = zR(a(w)L) - wL = zR(a(w)L) - \frac{w}{a(w)} a(w)L$$

No matter what the optimal level of  $a(w)L$  is, set  $w/a(w)$  as small as possible.

- Numerator in  $w/a(w)$ : the price for one hour of work  
Denominator: efficiency units per hour  
The fraction  $w/a(w)$ : price per efficiency unit. This should be minimized.
- To minimize  $w/a(w)$  requires that an increase of 1% in  $w$  increases  $a$  by 1% (if  $a$  were increased by more,  $w/a(w)$  would decrease, etc.). That is:

$$\frac{da/a}{dw/w} = \frac{a'(w)w}{a(w)} = 1.$$

- Is the unconstrained optimum,  $w^*$ , interior, and hence given by the first order conditions? This depends on the shapes of the involved curves. A typical efficiency curve could look like:



- In this case the optimal wage rate,  $w^*$ , is interior.

- We can insert the optimal wage rate,  $w^*$ , into one of our first order conditions to find:

$$zR'(a(w^*)L)a(w^*) = w^*.$$

- The solution  $L^*$  is interior ( $0 < L^* < \infty$ ) if, e.g.,  $R'(0) = \infty$  and  $R'(\infty) = 0$ . The unconstrained optimum,  $(w^*, L^*)$ , is then interior.
- If furthermore  $L^* < L^s(w^*)$ , then  $(w^*, L^*)$  is also the constrained optimum and at this optimum there is **unemployment and no tendency to lower real wages.**



- It was only for a “clean argument” we assumed that the firm was the sole buyer in the labour market. What if there are many buying firms?
- Under traditional circumstances (without efficiency wages) this leads to wage taking behaviour. An informal argument for this goes:
  - At the going market wage each firm can get all the labour it wants.
  - Should a firm set a lower wage? Workers would seek the other firms and these would be able to absorb the additional labour because they are many. So: no!
  - Should it set a higher wage? The firm would be overflowed with labour, but it would have no interest in this since it is already able to get all the labour it wants at the lower wage. So: no!

- Efficiency wages can give the firm an interest in setting the wage above the level required for getting the labour the firm wants and nothing prevents it from doing so. The firm just abstains from hiring all the labour in supply at the wage it sets.
- If there are  $n$  firms, each can have its own unconstrained optimum,  $(w^*, L^*)$ . Hence, unconstrained labour demand is  $nL^*$ .
- If  $nL^* < L^s(w^*)$ , there is **unemployment at fully adjusted real wages**.
- Whether there are many or few firms in the partial equilibrium model: Equilibrium with unemployment is a **possibility** (but not a necessity).

# MODEL IMPLICATIONS

1. Possibility of **truly involuntary unemployment in an equilibrium with a fully adjusted real wage**
  - Employers (not employees) drive up wages to attain an optimal level of productivity from each hired worker, and the wages thus determined (by the employers) may be so high that the employers do not want to hire all the workers who want jobs at those wages.
2. Shifts in the supply or demand shock  $z$  do not affect the real wage,  $w^*$ , but do affect employment,  $L^*$ .
  - Fits well with an important stylized fact of the business cycle: over the cycle, employment is much more positively correlated with GDP than real wages are.
  - Does not fit with an important fact of growth: in the long run, real wages increase with productivity.
  - Is the latter a problem for the efficiency wages model? No! (Exercise 6)

# THE CAUSES OF EFFICIENCY WAGES

The effectivity of the individual worker or work hour depends positively on the real wage, e.g.,

$$a = a(w) = \tilde{a}(w - v) \quad \text{where } v \geq 0, \tilde{a}(0) = 0 \text{ and } \tilde{a}' > 0$$

- 0. The health argument (probably not very important in developed countries)
  - 1. Keeping the workers in the firm
  - 2. Getting good workers at hirings
  - 3. The shirking argument
  - 4. Reciprocity: treating others like they treat you

We take a closer look at...

# THE SHIRKING MODEL

- A firm has a certain number of employees. For each worker, if an effort level of  $a \geq 0$  is exerted and a real wage rate of  $w$  is received, then the utility obtained is

$$w - c(a)$$

$$\text{where } c(0) = 0, c' > 0, c'' > 0, c'(0) = 0$$

- If a worker is laid off, the utility consequence is  $v$ , an exogenous "outside option" or "normal income".
- The firm offers each worker a contract  $(w, \hat{a})$ , with wage rate  $w$ , and required effort  $\hat{a}$ .
- A worker exerting  $a < \hat{a}$  is shirking.
- Shirking is discovered with probability  $q$ ,  $0 < q < 1$ , independently for each worker. The consequence of discovered shirking is that the worker is fired.

- For  $a < \hat{a}$  (shirking) expected utility is  $(1 - q)[w - c(a)] + qv$ . Might as well set  $a = 0 \Rightarrow \text{utility} = (1 - q)w + qv$ .
- For  $a \geq \hat{a}$  (no-shirking) expected utility is  $w - c(a)$ . Might as well set  $a = \hat{a} \Rightarrow \text{utility} = w - c(\hat{a})$ .
- The no-shirking condition is:

$$w - c(\hat{a}) \geq (1 - q)w + qv \Leftrightarrow w \geq v + \frac{c(\hat{a})}{q}.$$

- A higher  $w$  permits a higher  $\hat{a}$ . A higher  $\hat{a}$  requires a higher  $w$ .
- Equating the right-hand side and the left-hand side of the no-shirking condition, and assuming that

$$c(a) = \beta a^{1/\eta}, \quad \beta > 0, \quad 0 < \eta < 1$$

gives

$$a = \hat{a}(w, v, q) = \left[ (q / \beta)(w - v) \right]^\eta$$

- "Typical" efficiency function:

$$a(w) = k(w - v)^\eta.$$

# THE WAGE CURVE

- The normal income/outside option must be

$$v = ub + (1 - u)w,$$

where  $w$  is the average wage level,  $u$  is the unemployment rate and  $b$  is the unemployment benefit interpreted to include the value of leisure and home production.

- Since all the firms face the same  $v$ , they all set the same  $w_i$ . Inserting this into the optimal wage (Solow) condition gives:

$$w = \frac{v}{1 - \eta} = \frac{ub + (1 - u)w}{1 - \eta} \Leftrightarrow$$

$$w = \frac{u}{u - \eta} b = \frac{1}{1 - \eta / u} b > b$$

(  $u > \eta$  will be a property of equilibrium, see next slide)

- The optimal real wage **at the aggregate level** is a mark-up over the benefit rate  $b$ , and the mark-up is increasing in  $\eta$  and decreasing in  $u$ .

# THE UNEMPLOYMENT RATE

- The normal income/outside option

$$v = ub + (1 - u)w,$$

can be solved for the unemployment rate, getting:

- $u = \frac{w}{w-b} \eta > \eta > 0$
- Hence the equilibrium unemployment rate is positive.



# CONCLUSIONS WRT. THE OCCURRENCE OF UNEMPLOYMENT

- We can show that in the macroeconomic equilibrium, where all prices and wages are fully adjusted, there is involuntary unemployment. We here give the intuition.
- In the partial equilibrium model involuntary unemployment is a possibility. Why is macro predicting its necessity?
- Remember that  $v = ub + (1 - u)w$ , so  $u = 0$  implies that  $v = w$ .
- Each firm wants to drive wages above  $v$ , which is equal to the wage level when  $u = 0$ . As long as  $u = 0$ , there is an upward pressure on  $w$ . Only when  $u > 0$ , one can have  $w > v$  as required for equilibrium.
- In essence, the firms push wages up (to stimulate productivity) to a level where they don't want to hire all the labour supplied.